Unit-II: Network Access &Tools

- **What is Ethernet?**

A local Area Network (LAN) is a data communication network connecting various terminals or computers within a building or limited geographical area. The connection between the devices could be wired or wireless. Ethernet, Token rings, and Wireless LAN using IEEE 802.11 are examples of standard LAN technologies.

What is Ethernet?

Ethernet is the most widely used LAN technology and is defined under IEEE standards 802.3. The reason behind its wide usability is that Ethernet is easy to understand, implement, and maintain, and allows low-cost network implementation. Also, Ethernet offers flexibility in terms of the topologies that are allowed. Ethernet generally uses a bus topology. Ethernet operates in two layers of the OSI model, the physical layer and the data link layer. For Ethernet, the protocol data unit is a frame since we mainly deal with DLLs. In order to handle collisions, the Access control mechanism used in Ethernet is CSMA/CD. Although Ethernet has been largely replaced by wireless networks, wired networking still uses Ethernet more frequently. Wi-Fi eliminates the need for cables by enabling users to connect their smartphones or laptops to a network wirelessly. The 802.11ac Wi-Fi standard offers faster maximum data transfer rates when compared to Gigabit Ethernet. However, wired connections are more secure and less susceptible to interference than wireless networks. This is the main justification for why so many companies and organizations continue to use Ethernet.

There are different types of Ethernet networks that are used to connect devices and transfer data.

Let's discuss them in simple terms:

1. Fast Ethernet: This type of Ethernet network uses cables called twisted pair or CAT5. It can transfer data at a speed of around 100 Mbps (megabits per second). Fast Ethernet uses both fiber optic and twisted pair cables to enable communication. There are three categories of Fast Ethernet: 100BASE-TX, 100BASE-FX, and 100BASE-T4.

2. Gigabit Ethernet: This is an upgrade from Fast Ethernet and is more common nowadays. It can transfer data at a speed of 1000 Mbps or 1 Gbps (gigabit per second). Gigabit Ethernet also uses fiber optic and twisted pair cables for communication. It often uses advanced cables like CAT5e, which can transfer data at a speed of 10 Gbps.

3.10-Gigabit Ethernet: This is an advanced and high-speed network that can transmit data at a speed of 10 gigabits per second. It uses special cables like CAT6a or CAT7 twisted-pair cables and fiber optic cables. With the help of fiber optic cables, this network can cover longer distances, up to around 10,000 meters.

4. Switch Ethernet: This type of network involves using switches or hubs to improve network performance. Each workstation in this network has its own dedicated connection, which improves the speed and efficiency of data transfer. Switch Ethernet supports a wide range of speeds, from 10 Mbps to 10 Gbps, depending on the version of Ethernet being used.

In summary, Fast Ethernet is the basic version with a speed of 100 Mbps, Gigabit Ethernet is faster with a speed of 1 Gbps, 10-Gigabit Ethernet is even faster with a speed of 10 Gbps, and Switch Ethernet uses switches or hubs to enhance network performance.

Key Features of Ethernet

1. **Speed:** Ethernet is capable of transmitting data at high speeds, with current Ethernet standards supporting speeds of up to 100 Gbps.
2. **Flexibility:** Ethernet is a flexible technology that can be used with a wide range of devices and operating systems. It can also be easily scaled to accommodate a growing number of users and devices.
3. **Reliability:** Ethernet is a reliable technology that uses error-correction techniques to ensure that data is transmitted accurately and efficiently.
4. **Cost-effectiveness:** Ethernet is a cost-effective technology that is widely available and easy to implement. It is also relatively low-maintenance, requiring minimal ongoing support.
5. **Interoperability:** Ethernet is an interoperable technology that allows devices from different manufacturers to communicate with each other seamlessly.
6. **Security:** Ethernet includes built-in security features, including encryption and authentication, to protect data from unauthorized access.
7. **Manageability:** Ethernet networks are easily managed, with various tools available to help network administrators monitor and control network traffic.
8. **Compatibility:** Ethernet is compatible with a wide range of other networking technologies, making it easy to integrate with other systems and devices.
9. **Availability:** Ethernet is a widely available technology that can be used in almost any setting, from homes and small offices to large data centers and enterprise-level networks.
10. **Simplicity:** Ethernet is a simple technology that is easy to understand and use. It does not require specialized knowledge or expertise to set up and configure, making it accessible to a wide range of users.
11. **Standardization:** Ethernet is a standardized technology, which means that all Ethernet devices and systems are designed to work together seamlessly. This makes it easier for network administrators to manage and troubleshoot Ethernet networks.
12. **Scalability:** Ethernet is highly scalable, which means it can easily accommodate the addition of new devices, users, and applications without sacrificing performance or reliability.
13. **Broad compatibility:** Ethernet is compatible with a wide range of protocols and technologies, including TCP/IP, HTTP, FTP, and others. This makes it a versatile technology that can be used in a variety of settings and applications.
14. **Ease of integration:** Ethernet can be easily integrated with other networking technologies, such as Wi-Fi and Bluetooth, to create a seamless and integrated network environment.
15. **Ease of troubleshooting:** Ethernet networks are easy to troubleshoot and diagnose, thanks to a range of built-in diagnostic and monitoring tools. This makes it easier for network administrators to identify and resolve issues quickly and efficiently.
16. **Support for multimedia:** Ethernet supports multimedia applications, such as video and audio streaming, making it ideal for use in settings where multimedia content is a key part of the user experience. Ethernet is a reliable, cost-effective, and widely used LAN technology that offers high-speed connectivity and easy manageability for local networks.

Advantages of Ethernet

Speed: When compared to a wireless connection, Ethernet provides significantly more speed. Because Ethernet is a one-to-one connection, this is the case. As a result, speeds of up to 10 Gigabits per second (Gbps) or even 100 Gigabits per second (Gbps) are possible.

Efficiency: An Ethernet cable, such as Cat6, consumes less electricity, even less than a wifi connection. As a result, these ethernet cables are thought to be the most energy-efficient.

Good data transfer quality: Because it is resistant to noise, the information transferred is of high quality.

Baud rate = 2* Bit rate

Ethernet LANs consist of network nodes and interconnecting media, or links.

The network nodes can be of two types:

Data Terminal Equipment (DTE): Media, Generally, DTEs are the end devices that convert the user information into signals or reconvert the received signals. DTE devices are: personal computers, workstations, file servers or print servers also referred to as end stations. These devices are either the source or the destination of data frames. The data terminal equipment may be a single piece of equipment or multiple pieces of equipment that are interconnected and perform all the required functions to allow the user to communicate. A user can interact with DTE or DTE may be a user.

Data Communication Equipment (DCE): DCEs are the intermediate network devices that receive and forward frames across the network. They may be either standalone devices such as repeaters, network switches, or routers, or maybe communications interface units such as interface cards and modems. The DCE performs functions such as signal conversion, coding, and maybe a part of the DTE or intermediate equipment.

Disadvantages of Ethernet

Distance limitations: Ethernet has distance limitations, with the maximum cable length for a standard Ethernet network being 100 meters. This means that it may not be suitable for larger networks that require longer distances.

Bandwidth sharing: Ethernet networks share bandwidth among all connected devices, which can result in reduced network speeds as the number of devices increases.

Security vulnerabilities: Although Ethernet includes built-in security features, it is still vulnerable to security breaches, including unauthorized access and data interception.

Complexity: Ethernet networks can be complex to set up and maintain, requiring specialized knowledge and expertise.

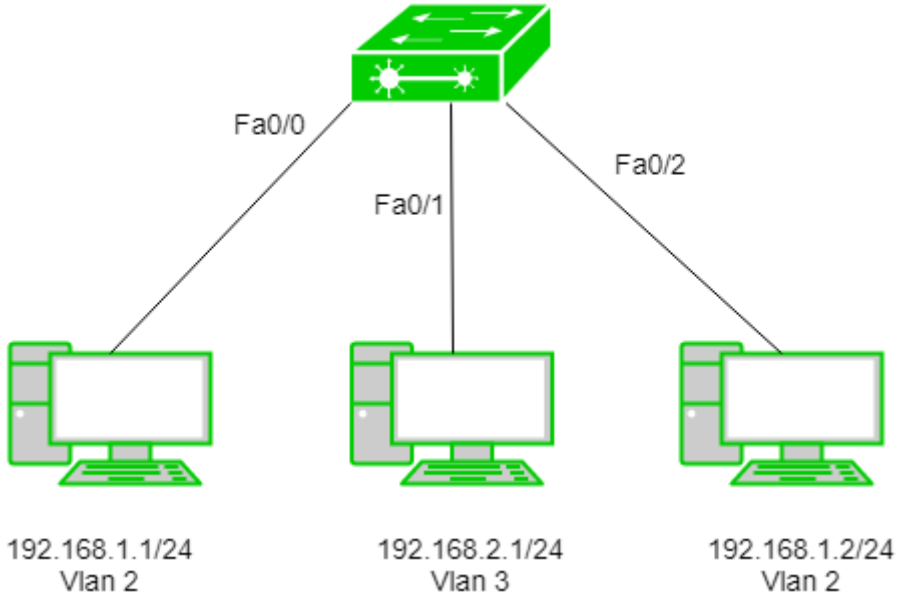
Compatibility issues: While Ethernet is generally interoperable with other networking technologies, compatibility issues can arise when integrating with older or legacy systems.

Cable installation: Ethernet networks require the installation of physical cables, which can be time-consuming and expensive to install.

Physical limitations: Ethernet networks require physical connections between devices, which can limit mobility and flexibility in network design.

How LAN Switches Work?

Switches are an important component of many networks because they speed up operations. Switches allow smooth and efficient direct communication between different nodes (network connection points, usually computers) on a network. Networks and switches come in many forms. A LAN switch is a switch that provides unique connectivity to each node in an organization's internal network. Essentially, a LAN switch sets up a series of temporary networks that span only the two devices currently exchanging data. In the Open Systems Interconnection (OSI) architecture, the Layer 2 data link layer is where network switches physically operate. It receives packets sent by a device connected to a physical port and sends them to their destination. Switches also work at layer 3 or the network layer where routing takes place. Networks based on Ethernet, Fiber Channel, Asynchronous Transfer Mode (ATM), and InfiniBand often include switches as components. Ethernet is used in most switches today.



Types of Switches:

- **Unmanaged Switch:** Unmanaged network switches are built into plug-and-play devices that require no configuration to function. Unmanaged switches are typically used for simple connectivity. If you are at work, in a lab, in a conference room, or just need a few extra ports, you will typically find them on your home network.
- **Managed Switch:** Managed switches can be configured to fit your network perfectly, increasing functionality, flexibility, and security. With enhanced management, you can protect your network more effectively and better serve your network users.
- **Network Switches and Hubs:** A local area network, also called a LAN, has a central connection point for devices called a network hub. Hub-based networks only allow users to share a fixed amount of bandwidth. As more devices connect to a network hub, data transfer times increase. Switches work around these and other network hub limitations. A large network may have multiple switches connected to multiple computer system groups. Routers typically connect to these switches and allow connected devices to access the Internet.

Working on Switch:

- The switch records the media access control (MAC) addresses of attached devices, a code embedded in the attached device's network interface card (NIC). The NIC is connected to an Ethernet cable connected to the switch. The switch uses MAC addresses to determine which devices to send outgoing packets to and where to send incoming packets.
- Network layers (Layer 3) IP addresses can be dynamically assigned to devices and can change over time, whereas MAC addresses identify physical devices and are static. (Think of the IP address as the license plate and the MAC address as his VIN on the car.)
- The switch scans the header of each packet that arrives, compares the destination address, and then delivers the packet to the destination device through the correct port. Most switches have the full-duplex capability, allowing packets sent to and received from a device to access the full bandwidth of the switch link, allowing network traffic to flow simultaneously to and from the switch and attached devices. less likely to collide. (Imagine two people having a conversation using a smartphone instead of a walkie-talkie.)
- The switch can operate on Layer 3 in addition to Layer 2. This is required to support virtual LANs (VLANs), logical network segments that can span subnets. The switch operates at Layer 2, but can also operate on Layer 3. The switch can operate on Layer 3 in addition to Layer 2. This is required to support virtual LANs (VLANs), logical network segments that can span subnets. The switch operates on Layer 2, but can also operate on Layer 3. Traffic must transit through switches in order to move from one subnet to another, and this is made easier by the switches' built-in routing capabilities.

Benefits of LAN Switch:

- **Reduced congestion and transmission delays:** Network segmentation reduces the difficulty users have in accessing the network. The available LAN bandwidth for each segment reduces this. Business applications improve access efficiency.
- **Increased scalability:** The network can grow independently according to the capacity of your business.
- **No Network Failures:** A well-designed network has a very low probability of network failures and users will always have a seamless network connection.
- **Bandwidth Performance:** Bandwidth performance in network environments has been improved for each user. Users can often interact with client-her-server databases or use applications such as multimedia. There is no delay when they communicate.
- **Parallel connection:** This allows different types of data to be transferred simultaneously between paired devices connected through a switch. This feature is not available if your connection uses a hub-based network.
- **Traceable and Secure with VLANs:** VLANs help maintain network connectivity. Maintain functionality without reconfiguration, even when the environment changes. Individual users can easily manage workgroup connections, protect their relationships, and conduct commercial activities based on common interests. This allows users to move more freely and enjoy the same connection as others.

VLAN Trunking Protocol (VTP)

To carry traffic of a VLAN, it must be first configured on the switch. Suppose, if the user wants to send a frame from source to destination and the shortest path between them contains 1000 switches. To process a frame of any VLAN, VLANs should be configured first so, have to configure the same VLANs on all the 1000 switches manually. It will not be possible for the administrator to do that. Here comes VTP to the rescue.

VLAN Trunking Protocol (VTP) – VTP is CISCO proprietary protocol used to maintain consistency throughout the network or the user can say that synchronizing the VLAN information in the same VTP domain. VTP allows you to add, delete and rename VLANs which is then propagated to other switches in the VTP domain. VTP advertisements can be sent over 802.1Q, and ISL trunks.

Requirements – There are some requirements for VTP to communicate VLAN information between switches. These are:

1. The VTP version must be same on the switches user wants to configure
2. VTP domain name must be same on the switches
3. One of the switches must be a server
4. Authentication should match if applied

VTP modes – There are 3 modes:

- **Server –** The switches are set to this mode by default. This mode allows you to create, add and delete VLANs. The changes you want to make should be done in this mode. Any changes that are done on this mode(on a particular switch) will be advertised to all the switches that are in the same VTP domain. In this mode, the configuration are saved in NVRAM.
- **Configuration –** User will first make the switch VTP server

```
Switch# config terminal

Switch(config)#vtp mode server
```

Now, User has to make a VTP domain and assign a password for authentication.

```
Switch(config)#vtp domain geeksforgeeks

Switch(config)#vtp password hardwork
```

User can verify the configuration by:

```
Switch(config)#do should vtp password

Switch(config)#do show vtp
```

- **Client –** In this mode, the switches receive the updates and can also forward the updates to other switches (which are in the same VTP domain). The updates received here are not saved in NVRAM so all the configuration will be deleted if the switch is reset or reloaded i.e the switches will only learn and pass the VTP summary advertisements to the other switches.
- **Configuration –** As the switches are set to server mode by default, therefore user can change it to client mode by:

```
Switch(config)#vtp mode client
```

- **Transparent –** This mode only forwards the VTP summary advertisements through trunk link. The transparent mode switches can make their own local database which keep secret from other switches. The whole purpose of transparent mode is to forward the VTP summary advertisements but not to take part in the VLAN assignments.
- **Configuration –** User can change the mode to transparent by

```
Switch(config)#vtp mode transparent
```

Configuration Revision Number – The configuration revision number is a 32-bit number that indicates the level of revision for a VTP packet. This configuration number is tracked by every switch in order to find that the received information is more recent than the current version. Every time one modification is done on the VLANs by the server switch, and the configuration revision number increases by one. The client mode devices receive it and check if the configuration revision number that they received is latest or not by comparing their own configuration number with the number received. If the configuration number is greater than their own number then the devices update their configuration and pass it to other clients of the same VTP domain. If the configuration number is the same then the devices just pass it to other clients of the same VTP domain. User can check the configuration revision number by:

```
switch(config)#do show vtp status
```

Use VTP in an Organization: As a matter of course, all switches are designed to be VTP servers. This design is appropriate for limited-scope networks in which the size of the VLAN data is little and the data is easily put away in all switches (in NVRAM). In an enormous organization, the organization chairman should settle on a decision sooner or later, when the NVRAM stockpiling that is important is inefficient in light of the fact that it is copied on each switch. Right now, the organization overseer should pick a couple of exceptional switches and keep them as VTP servers. All the other things that partake in VTP can be transformed into a client. The quantity of VTP servers should be picked to give the level of overt repetitiveness that is wanted in the organization.

❖ WLAN

WLAN stands for **Wireless Local Area Network.** WLAN is a local area network that uses radio communication to provide mobility to the network users while maintaining the connectivity to the wired network. A WLAN basically, extends a wired local area network. WLAN's are built by attaching a device called the access point(AP) to the edge of the wired network. Clients communicate with the AP using a wireless network adapter which is similar in function to an ethernet adapter. It is also called a LAWN is a Local area wireless network.

The performance of WLAN is high compared to other wireless networks. The coverage of WLAN is within a campus or building or that tech park. It is used in the mobile propagation of wired networks. The standards of WLAN are Hiper LAN, Wi-Fi, and IEEE 802.11. It offers service to the desktop laptop, mobile application, and all the devices that work on the Internet. WLAN is an affordable method and can be set up in 24 hours. WLAN gives users the mobility to move around within a local coverage area and still be connected to the network. Most latest brands are based on IEE 802.11 standards, which are the WI-FI brand name.

History

A professor at the University of Hawaii who's name was Norman Abramson, developed the world's first wireless computer communication network. In 1979, Gfeller and u. Bapst published a paper in the IEE proceedings reporting an experimental wireless local area network using diffused infrared communications. The first of the IEEE workshops on Wireless LAN was held in 1991.

WLAN Architecture

Components in Wireless LAN architecture as per IEEE standards are as follows:

1. **Stations:** Stations consist of all the equipment that is used to connect all wireless LANs. Each station has a wireless network controller.
2. **Base Service Set(BSS):** It is a group of stations communicating at the physical layer.
3. **Extended Service Set(ESS):** It is a group of connected Base Service Set(BSS).
4. **Distribution Service (DS):** It connects all Extended Service Set(ESS).

Types of WLANs

As per IEEE standard WLAN is categorized into two basic modes, which are as follows:

1. **Infrastructure:** In Infrastructure mode, all the endpoints are connected to a base station and communicate through that; and this can also enable internet access. A WLAN infrastructure can be set up with: a wireless router (base station) and an endpoint (computer, mobile phone, etc). An office or home WiFi connection is an example of Infrastructure mode.
2. **Ad Hoc:** In Ad Hoc mode WLAN connects devices without a base station, like a computer workstation. An Ad Hoc WLAN is easy to set up it provides peer-to-peer communication. It requires two or more endpoints with built-in radio transmission.

Working of WLAN

WLAN transmits data over radio signals and the data is sent in the form of a packet. Each packet consists of layers, labels, and instructions with unique MAC addresses assigned to endpoints. This enables routing data packets to correct locations.

Characteristics of WLAN

1. Seamless operation.
2. Low power for battery use.
3. Simple management, easy to use for everyone.
4. Protection of investment in wired networks.
5. Robust transmission technology.

Advantages of WLAN

1. Installation speed and simplicity.
2. Installation flexibility.
3. Reduced cost of ownership.
4. Reliability.
5. Mobility.
6. Robustness.

Disadvantages of WLAN

1. Slower bandwidth.
2. Security for wireless LANs is the prime concern.
3. Less capacity.
4. Wireless networks cost four times more than wired network cards.
5. Wireless devices emit low levels of RF which can be harmful to our health.

What is Network Access Control?

Network Access Control is a security solution that uses a set of protocols to keep unauthorized users and devices out of a private network or give restricted access to the devices which are compliant with network security policies. It is also known as **Network Admission Control.** It handles network management and security that implements security policy, compliance, and management of access control to a network.

NAC works on wired and wireless networks by identifying different devices that are connected to the network. For setting up an NAC network security solution, administrators will determine the protocols that will decide how devices and users are authorized for the right level of authorization. Access rules are generally based on the criterion such as device used, the location accessed from, the access rights of various individuals, as well as the specific data and resources being accessed.

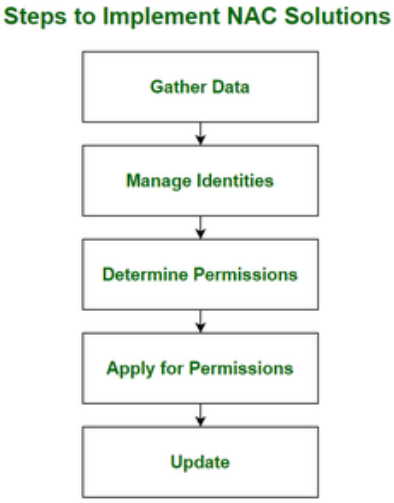
Components of Network Access Control Scheme:

1. **Restricted Access:** It restricts access to the network by user authentication and authorization control. For example, the user can't access a protected network resource without permission to access it.
2. **Network Boundary Protection:** It monitors and controls the connectivity of networks with external networks. It includes tools such as controlled interfaces, intrusion detection, and anti-virus tools. It is also called perimeter defense. For example, the firewall can be used to prevent unauthorized access to network resources from outside of the network.

Types of Network Access Control:

1. **Pre-admission:** It happens before access to the network is granted on initialization of request by user or device to access the network. It evaluates the access attempt and only allows the access if the user or device is compliant with organization security policies and authorized to access the network.
2. **Post-admission:** It happens within the network when the user or device attempts to access the different parts of the network. It restricts the lateral movement of the device within the network by asking for re-authentication for each request to access a different part of the network.

Steps to Implement NAC Solutions:



Implement NAC Solutions

1. **Gather Data:** Perform an exhaustive survey and collect information about every device, user, and server that has to interface with the network resources.
2. **Manage Identities:** Verify user identities within the organization by authentication and authorization.
3. **Determine Permissions:** Create permission policies stating different access levels for identified user groups.
4. **Apply for Permissions:** Apply permission policies on identified user groups and register each user in the NAC system to trace their access level and activity within the network.
5. **Update:** Monitor security operations and make adjustments to permission policies based on changing requirements of the organization with time.

Importance of Network Access Control:

There has been exponential growth in the number of mobile devices accessing private networks of organizations in the past few years. This has led to an increase in security risks for the organization's resources and therefore, some tools are required that can provide the visibility, access control, and compliance capabilities to strengthen the network security infrastructure.

A NAC system can deny network access to non-compliant devices or give them only restricted access to computing resources, thus preventing insecure nodes from infecting the network. Also, NAC products can handle large enterprise networks that have a large range of different device types connected to the network.

Responsibilities:

1. It allows only compliant, authenticated devices to access network resources and infrastructure.
2. It controls and monitors the activity of connected devices on the network.
3. It restricts the availability of network resources of private organizations to devices that follow their security policy.
4. It regulates the access of network resources to the users.
5. It mitigates network threats by enforcing security policies that block, isolate, and repair non-compliant machines without administrator attention.

Common Use-Cases:

1. Organizations that allow employees to use their own devices or take corporate devices home use NAC to ensure network security.
2. Organizations use NAC to grant access to different network resources to people or devices that are outside of the organization and are subjected to different security controls.
3. NAC protects from threats caused due to use of IoT devices by categorizing IoT devices into groups that have limited permission and constantly monitoring their activities.

Benefits:

1. Users can be required to authenticate via multi-factor authentication, which is much more secure than identifying users based on IP addresses or username and password combinations.
2. It provides additional levels of protection around individual parts of the network.

Limitations:

1. It has low visibility in IoT devices and devices with no specific users associated with it.
2. It does not protect from threats present inside the network.

3. It may not work for organizations if it is not compatible with existing security controls.

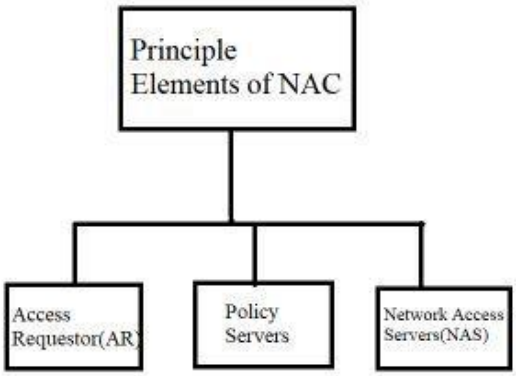
Principle Elements of NAC (Network Access Control):

There are mainly three principle elements of NAC which are:

1.Access Requestor(AR).

2.Policy Servers.

3.Network Access Servers(NAS).



Three Principle Elements of NAC (Network Access Control).

Let's look at them one by one now:

1.Access Requestor(AR): We may determine from the name that it is someone attempting to gain access by requesting it. This access can be granted to any entity, such as a device, person, or process.

This entity attempts to get access to network resources. It might be any device handled by the NAC system, such as servers, cameras, printers, and other IP-enabled devices.

ARs are also known as supplicants or clients at times. ARs ensures that no entity has illegal access to protected resources.

To get access, these ARs must follow to the organization's specific guidelines or policies.

2.Policy Server: The policy server analyzes what access should be provided to AR based on the AR's identity, permission level, attempted request, and an organization's established access policy.

The policy server frequently relies on backend services, such as antivirus, patch management, or a user directory, to function.

The policy server helps to determine the host's state. An organization creates different access policies to clearly authorize or reject such access. If the AR follows the organization's policy, the policy server gives access based on the requestor's permission; otherwise, the AR will not be permitted access based on its permission.

These tools offer highly detailed ways to set organizational rules and control the organization's full IP infrastructure.

3.Network Access Server(NAS): Users connecting to an organization's internal network from distant locations utilize the NAS as an access control point. These often serve as VPNs and give users access to the company's internal network. These days, NAS functionality is frequently included in policy server systems.

❖ Basic Device Configuration

What is Network Configuration?

Network configuration refers to the organization, governance, and maintenance of a network with the right set of controls and policies.

Network Configuration Definition

Network configuration focuses on managing a network and its devices by applying the right set of policies, controls, and configurations. It encompasses activities from device discovery to configuration backups for efficient network administration.

What are the types of network configuration?

Network administrators maintain a well-organized information repository of the network devices with details, such as device location or network address and device settings, as part of configuration management. This configuration database works as a guiding source for admins while making updates and changes in the network.

Generally, network topologies denote various types of network configuration. Network topology refers to the systematic arrangement of nodes or devices in a network that allows them to exchange information.

Network topologies are of two types—physical and logical. Physical topology depicts the linkages between physical devices via cables, wires, etc., in a network. In contrast, logical topology denotes how information is transferred through a network. The way devices interact in a network is also a part of the logical topology.

Some of the popular physical network topologies are as follows:

- **Bus:** Every node or device in the network is connected in a linear order with a unidirectional data flow. Bus topology is cost-effective, but it can break down quickly when there's high network traffic.
- **Ring:** Nodes are connected circularly, while data can flow in one or both directions as per needs. Ring networks are easy to set up and expand, but troubleshooting is often challenging.
- **Star:** A central server or node manages all other nodes with point-to-point communication. Star topology is commonly used in local area networks because of

benefits such as centralized control, better security, and easy configuration. However, the entire network can crumble if the central server fails.

- **Mesh:** Nodes are linked in a web-like structure with point-to-point connections with every other node in the network. Data transmits through routing (shortest-path approach) and flooding methods (broadcast approach). Mesh networks are highly reliable but expensive to set up and maintain.
- **Tree:** Nodes are interconnected in hierarchical order with at least three levels. Tree network is an extension of star topology and is used in wide area networks (WANs).
- **Hybrid:** Hybrid combines two or more topologies. Organizations looking for flexibility in their IT infrastructure prefer hybrid networks.

Network configuration key areas

Device discovery and management: Configuration management discovers devices in your network and stores their critical details such as port configuration and interface details in a network inventory. Having up-to-date inventory also helps you generate reports quickly using filters such as device type, device vendor, and device location. Centralized inventory makes management easy.

Configuration backup and restore: A network disaster can happen at any time, so an organization should regularly take network backups to stay future-ready. Regular backups allow you to quickly recover from unexpected scenarios such as equipment failure by uploading the most stable versions of your device settings.

Change management: Maintaining accurate change records helps you identify the network engineers behind a particular configuration modification in your network. While this doesn't necessarily reduce network problems, it can accelerate the error resolution process. Standardization of change management practices, a crucial part of configuration management, ensures modifications across the network are well documented, prompt, and traceable.

Policy management and compliance audits: Before making configuration changes, network administrators must formulate robust network policies to demonstrate compliance with various security standards such as PCI DSS and DISA STIG. Network security policy creation and regular compliance audits of device settings are vital in configuration management.

Network automation: Automation of network administration tasks such as device backups, firmware updates, and access rights modification is among the configuration management best practices. This boosts your network teams' productivity in addition to eliminating repetitive tasks and human errors.

❖ Network Tools:

1. OPNET

OPNET Network simulator is a tool to simulate the behavior and performance of any type of network. The main difference Opnet Network Simulator comparing to other simulators lies in its power and versatility. IT Guru provides pre-built models of protocols and devices. It allows you to create and simulation different network topologies. The set of protocols/devices is fixed – you cannot create new protocols nor modify the behavior of existing ones.
The OPNET network simulator is a tool for simulating the behavior and performance of any type of network used by online casino players. The main difference between Opnet Network Simulator and other simulators is its power and versatility, read also about Kirk Cousins brings out inner Paul Wall: goes topless and 'ices out' after Washington revenge win. Opnet Network Simulator is also free software, which will definitely be a plus for many online casino players.

Advantages of Opnet Network Simulator:

- Opnet Network Simulator is a open free software
- Large number of project scenarios that are offered information on Opnet Network Simulator
- Can be overlooked using Opnet Network Simulator.

Uses of opnet simulator:

- Operational validation.
- Application troubleshooting.
- Network planning and design.
- Validating hardware architecture.
- Protocol modeling.
- Traffic modeling of telecommunication networks.
- Evaluating performance aspects of complex software systems.

2. IPERF

IPERF is a tool for network performance measurement and tuning. It is a cross-platform tool that can produce standardized performance measurements for any network. iperf has client and server functionality, and can create data streams to measure the throughput between the two ends in one or both directions.[2] Typical iperf output contains a time-stamped report of the amount of data transferred and the throughput measured.

The data streams can be either Transmission Control Protocol (TCP) or User Datagram Protocol (UDP):

- *UDP*: When used for testing UDP capacity, iperf allows the user to specify the datagram size and provides results for the datagram throughput and the packet loss.
- *TCP*: When used for testing TCP capacity, iperf measures the throughput of the payload. iperf uses 1024 × 1024 for mebibytes and 1000 × 1000 for megabytes.

iperf is open-source software written in C, and it runs on various platforms including Linux, Unix and Windows (either natively or inside Cygwin[3]). The availability of the source code enables the user to scrutinize the measurement methodology.

iperf is a compatible reimplementation of the ttcp program that was developed at the National Center for Supercomputing Applications at the University of Illinois by the Distributed Applications Support Team (DAST) of the National Laboratory for Applied Network Research (NLANR),[4] which was shut down on December 31, 2006, on termination of funding by the United States National Science Foundation.

3. JPERF

JPerf is a graphical front end for the popular network testing tool Iperf. Using JPerf, you can quickly test a WAN or LAN connection to determine the maximum network throughput. The test results are automatically graphed and presented in a format that is easy to read. JPerf can also be used to detect packet loss, delay, jitter, and other common network problems.

JPerf provides many benefits over Iperf, which is a command line-only application. Besides being reliable and easy to use, JPerf is completely free. The utility is fully open source and runs on both Windows and Linux systems.

Whether you're trying to diagnose a problem in your home network or troubleshoot the performance of a WAN link, JPerf can handle the task.

What is Wireshark?

Wireshark is a network packet analyzer. A network packet analyzer presents captured packet data in as much detail as possible.

You could think of a network packet analyzer as a measuring device for examining what's happening inside a network cable, just like an electrician uses a voltmeter for examining what's happening inside an electric cable (but at a higher level, of course).

In the past, such tools were either very expensive, proprietary, or both. However, with the advent of Wireshark, that has changed. Wireshark is available for free, is open source, and is one of the best packet analyzers available today.

Some intended purposes

Here are some reasons people use Wireshark:

- Network administrators use it to *troubleshoot network problems*
- Network security engineers use it to *examine security problems*
- QA engineers use it to *verify network applications*
- Developers use it to *debug protocol implementations*
- People use it to *learn network protocol* internals

Wireshark can also be helpful in many other situations.

Features

The following are some of the many features Wireshark provides:

- Available for *UNIX* and *Windows*.
- *Capture* live packet data from a network interface.
- *Open* files containing packet data captured with tcpdump/WinDump, Wireshark, and many other packet capture programs.
- *Import* packets from text files containing hex dumps of packet data.
- Display packets with *very detailed protocol information*.
- *Save* packet data captured.
- *Export* some or all packets in a number of capture file formats.
- *Filter packets* on many criteria.
- *Search* for packets on many criteria.
- *Colorize* packet display based on filters.
- Create various *statistics*.

DummyNet Emulator

Dummynet is a feature of the ipfw firewall. It exposes three rule types that let you control how traffic is managed once it is fed into the emulator: pipes, schedulers, and queues. A pipe emulates a network link, you can set characteristics on a pipe and they will be enforced by the emulator. This allows you to limit the available bandwidth or impose additional delay on packets that traverse the pipe. Queues and scheduler rules allow us to use dummynet as a traffic shaper, packets fed into a queue are classified by their 5-tuple (protocol, source and destination IP and port) and scheduled differently based on the algorithm and parameters provided.
To start using dummynet you need to load the ipfw and dummynet kernel modules. You should always experiment with firewall rules on a test machine, ideally one where you can easily get access to the console when you lock yourself out.
When testing rules, it can be helpful to change the ipfw default rule to accept rather than reject all traffic. This helps avoid lock out and removes some of the more perplexing forwarding issues. You can do this before loading the kernel module with the kenv utility, or by adding a very late rule that allows all traffic through.